Automation Bioengineer

Oak Ridge National Laboratory · Oak Ridge, TN

Job ID (this archive)	ENERGY-086
Employer	Oak Ridge National Laboratory
Location(s)	Oak Ridge, TN
Career stage (archive tag)	Graduate degree / research (Advanced degree / Research)
Location group	Other U.S.
Majors (archive tags)	Materials Science & Engineering, Data Science, Biomedical Engineering
Energy area	Energy laboratory / bioengineering
Pay listed	Not listed
Recruiting status when checked	Status not confirmed: the posting text was found, but current hiring was not verified. Checked Oct 26, 2026.
Source page	https://jobs.ornl.gov/job/Oak-Ridge-Automation-Bioengineer-TN-37830/1443084800/
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-057

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Requisition Id 17317

Overview:

Oak Ridge National Laboratory (ORNL) is seeking a Research Professional to advance automation-enabled biological research across the Biosciences Division (BSD). The Automation Bioengineer will work with scientists, engineers, and facility staff to identify high-value research needs and develop, integrate, and implement automated and semi-automated workflows that improve the scale, reproducibility, speed, and quality of biological experimentation. The work will span BSD programs and biological systems, including microbial and plant engineering, phenotyping, molecular and cellular biology, analytical workflows, and other areas where automation can strengthen research. The position will help connect and extend capabilities in the Advanced Plant Phenotyping Laboratory (APPL), greenhouse and plant transformation environments, microbial engineering laboratories, and other relevant facilities. The successful candidate will prototype and test approaches, document and support workflows, and work with engineering, computing, and technical teams across ORNL to translate research needs into practical automation solutions. The candidate will collaborate across BSD and with partners elsewhere at ORNL, including the Environmental Sciences Division (ESD), as appropriate. This is a cross-cutting research and capability-development role; it is not limited to one organism, group, or facility. The candidate will contribute to proposals, reports, publications, and presentations that communicate results and expand use of automation-enabled research capabilities.

Major Duties/Responsibilities:

- Develop and coordinate a cross-cutting automation bioengineering effort that responds to research needs across BSD and supports collaboration with other ORNL divisions and external partners.
- Work with researchers to assess workflows and design, prototype, integrate, validate, and improve automation solutions for biological research, including liquid handling, sample preparation, transformation and engineering, cultivation, imaging and phenotyping, analytical measurements, and data capture, as appropriate to project needs.
- Connect and extend automation capabilities across microbial, plant, molecular, cellular, and other biological systems, coordinating with APPL, greenhouse and plant transformation operations, microbial engineering laboratories, and other relevant facilities.
- Collaborate with engineering, instrumentation, computing, data science, and facility teams to develop fit-for-purpose hardware, software, interfaces, data practices, and safe operating procedures; identify when existing systems can be

adapted and when new prototypes are warranted.

- Lead or contribute to research projects and proposals that advance automation-enabled biological discovery, predictive design, and experimental throughput; document outcomes and contribute to scientific publications, technical reports, and presentations.
- Train and mentor researchers, students, and technical staff in the use, maintenance, and appropriate adaptation of automated workflows; promote documentation and knowledge sharing so capabilities can be used and sustained across teams.
- Coordinate project planning and execution with investigators and facility staff, including defining requirements, milestones, resource needs, and measures of success.
- Maintain detailed and accurate experimental, equipment, software, and workflow records; support data quality, reproducibility, and appropriate integration with laboratory information and data systems.
- Support day-to-day laboratory and prototype operations, including equipment readiness, troubleshooting, and compliance with ORNL environmental, health, safety, and security requirements.
- Build effective working relationships across disciplines and organizational boundaries, communicate technical concepts clearly, and contribute to a respectful, collaborative research environment.
- Maintain a culture of continued learning and keep current with relevant developments in laboratory automation, bioengineering, instrumentation, and experimental methods.
- Deliver ORNL's mission by aligning behaviors, priorities, and interactions with our core values of Impact, Integrity, Teamwork, Safety, and Service. Promote equal opportunity by fostering a respectful workplace in how we treat one another, work together, and measure success.

Basic Qualifications:

- A Ph.D. in bioengineering, biological engineering, biomedical engineering, chemical engineering, biology, biochemistry, molecular biology, or a related field, with relevant experience in laboratory automation, automation engineering, or automation-enabled biological research; OR a master's degree in a relevant field with at least five years of directly relevant experience.
- Demonstrated ability to develop, integrate, or implement automated or semi-automated experimental workflows in a research or biotechnology setting.

Preferred Qualifications:

- Experience programming, integrating, and troubleshooting laboratory automation technologies, including liquid handlers, robotic platforms, instruments, and associated software (i.e. Tecan Fluent) and more modular technologies (i.e. Opentrons).
- Experience translating researcher needs into robust, documented, and maintainable experimental workflows across one or more biological systems.
- Experience with engineering non-model bacteria.
- Experience with automation in areas such as microbial or plant engineering, molecular and cellular biology, phenotyping, analytical workflows, or high-throughput experimentation.
- Experience working with engineers, instrumentation specialists, computing or data teams, and multidisciplinary research groups.
- Experience with experimental data capture, metadata practices, data quality, and analysis or visualization of automation-generated datasets.
- Demonstrated ability to plan and execute projects, manage priorities, and communicate technical work through presentations, reports, proposals, or publications.
- Experience training or mentoring users and supporting shared or facility-based research capabilities.
- Ability to work independently while building productive collaborations across organizational boundaries.
- Commitment to safe, reproducible, and inclusive research practices.
- Ability to function well in a fast-paced research environment, set priorities to accomplish multiple tasks within deadlines, and adapt to ever changing needs.

Special Requirements:

- For employment at Oak Ridge National Laboratory (ORNL), a Real ID compliant form of identification will be required. Additionally, ORNL is subject to Department of Energy (DOE) access restrictions. All employees must also be able to obtain and maintain a federal Personal Identity Verification (PIV) card as mandated by Homeland Security Presidential Directive 12 (HSPD-12) and Department of Energy (DOE) Order 473.1A, which requires a favorable post-employment background investigation.

- To obtain this credential, new employees must successfully complete and pass a Federal Tier 1 background check investigation. This investigation includes a declaration of illegal drug activities, including use, supply, possession, or manufacture within the last year. This includes marijuana and cannabis derivatives, which are still considered illegal under federal law, regardless of state laws.
- For foreign national candidates: If you have not resided in the U.S. for three consecutive years, you are not eligible for the PIV credential and instead will need to obtain a favorable Local Site Specific Only (LSSO) risk determination to maintain employment. Once you meet the three-year residency requirement, you will be required to obtain a PIV credential to maintain employment.

Please submit three letters of reference when applying to this position. You may upload these directly to your application or have them sent to ORNLRecruiting@ornl.gov with the position title and number referenced in the subject line.

Instructions to upload documents to your candidate profile:

- Login to your account via jobs.ornl.gov
- View Profile
- Under the My Documents section, select Add a Document

About ORNL:

As a U.S. Department of Energy (DOE) Office of Science national laboratory, ORNL has an impressive 80-year legacy of addressing the nation's most pressing challenges. Our team is made up of over 7,000 dedicated and innovative individuals! Our goal is to create an environment where a variety of perspectives and backgrounds are valued, ensuring ORNL is known as a top choice for employment. These principles are essential for supporting our broader mission to drive scientific breakthroughs and translate them into solutions for energy, environmental, and security challenges facing the nation.

ORNL offers competitive pay and benefits programs to attract and retain individuals who demonstrate exceptional work behaviors. The laboratory provides a range of employee benefits, including medical and retirement plans and flexible work hours, to support the well-being of you and your family. Employee amenities such as on-site fitness, banking, and cafeteria facilities are also available for added convenience.

Other benefits include the following: Prescription Drug Plan, Dental Plan, Vision Plan, 401(k) Retirement Plan, Contributory Pension Plan, Life Insurance, Disability Benefits, Generous Vacation and Holidays, Parental Leave, Legal Insurance with Identity Theft Protection, Employee Assistance Plan, Flexible Spending Accounts, Health Savings Accounts, Wellness Programs, Educational Assistance, Relocation Assistance, and Employee Discounts.

This position will remain open for a minimum of 5 days after which it will close when a qualified candidate is identified and/or hired.

We accept Word (.doc, .docx), Adobe (unsecured .pdf), Rich Text Format (.rtf), and HTML (.htm, .html) up to 5MB in size. Resumes from third party vendors will not be accepted; these resumes will be deleted and the candidates submitted will not be considered for employment.

If you have trouble applying for a position, please email ORNLRecruiting@ornl.gov.

ORNL is an equal opportunity employer. All qualified applicants, including individuals with disabilities and protected veterans, are encouraged to apply. UT-Battelle is an E-Verify employer.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting description retrieved; application acceptance not independently established. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: A Ph.D. in bioengineering, biological engineering, biomedical engineering, chemical engineering, biology, biochemistry, molecular biology, or a related field, with relevant experience in laboratory automation, automation engineering, or automation-enabled biological research; OR a master's degree in a relevant field with at least five years of directly relevant experience.

Key skills listed in the archive: automation

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Unverified · PDF: Complete · Source: Official employer careers Employer / ATS identifier: 17317

Career stage: Advanced degree / Research

Major fit: Direct degree match: Biomedical Engineering explicitly accepted; PhD or relevant master's plus five years required. Energy-sector connection: Automation of microbial and plant research at a DOE laboratory; broader biosciences role, not exclusively energy projects.

Retrieval note: New posting found in expansion round. Official posting description retrieved; application acceptance not independently established. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer careers. Source identifier: 17317.

Sources and retrieval details

Employer posting

https://jobs.ornl.gov/job/Oak-Ridge-Automation-Bioengineer-TN-37830/1443084800/

Retrieved source

https://jobs.ornl.gov/job/Oak-Ridge-Automation-Bioengineer-TN-37830/1443084800/

Retrieved: 2026-10-26T16:37:10.148348+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.